

MUSCULOSKELETAL INJURIES MODULE

What to remember

Online Module Quick Reference

Each lesson of the online module is distilled to its four or five essential ideas. This document does not replace the manual, the interactive capsules or in-class practice. It gives you a reference point before the in-person session and a quick refresher after the course.

Musculoskeletal injuries, especially of the lower limbs, are among the most frequent injuries in the backcountry. The online module gives you the reasoning: mechanism, assessment, decision. The in-class module gives you the hands-on skills: immobilization, one joint and one bone, with your own equipment.

MODULE CONTENTS: FOUR LESSONS

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01 The musculoskeletal system

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00

Introduction to the chapter

Three lessons to understand, decide and immobilize

Why this chapter: in the field, you will never see the bone. You will see pain, deformity, loss of function. Your job is to decide whether you can let the patient walk, whether to immobilize, and how fast to evacuate.

- 1 Four injuries to recognize.** Fracture (the bone), dislocation (the joint), sprain (the ligament), strain (the muscle or tendon). Their signs often look alike; when in doubt, treat as a fracture.
- 2 One rule before all others: CSM.** Circulation, sensation, motion distal to the injury, before and after every manoeuvre, then at regular intervals. Impaired CSM changes the evacuation priority.
- 3 RICE for almost everything.** Rest, ice, compression, elevation. This is the management common to all four injuries once the injury has been stabilized.
- 4 Reasoning online, hands-on in class.** The Field Manual (pages 128 to 154) is your reference tool for immobilization: look at the picture, reproduce it. In class, each participant immobilizes at least one bone and one joint with their own equipment.
- 5 Three scenarios in the in-person module.** Knee or ankle sprain, muscle strain, muscle cramps, in addition to the fracture. Prepare by reviewing the signs and management of each.

LEARNER'S MANUAL

Section 9, Musculoskeletal Injuries ·
Online encyclopedia

FIELD MANUAL

Pages 128 to 154

IN CLASS

Musculoskeletal system video, review of
PQRST and RICE, splinting workshop,
scenario

01

The musculoskeletal system

Understanding the structures to recognize what is injured

The locomotor system is the skeleton and the musculature: bones, joints, ligaments, cartilage, muscles and tendons. Together, they give the body its shape, protect the organs, and produce movement and heat.

Structure	Typical injury	What gives it away
Bone	Closed or open fracture	Localized pain, deformity, loss of function, crepitus
Joint	Dislocation	Deformity, locked joint, loss of mobility
Ligament	Sprain	Pain on movement, swelling, instability
Muscle and tendon	Strain, tendinitis, cramp	Pain on contraction, stiffness, burning at the start of exertion

Figure 1. Four structures, four typical injuries, and what gives them away on examination.

- 1 Bones: five types, two skeletons.** Long, short, flat, irregular, sesamoid. The axial skeleton protects (skull, spine, rib cage); the appendicular skeleton moves (limbs, girdles). It takes considerable force to fracture a bone: look for the mechanism.
- 2 Mobile joints: the most mobile are the least stable.** Cartilage, capsule, synovial fluid, ligaments. Shoulder, kneecap and fingers are the most frequent dislocations. Hinge (knee, elbow): a single plane; ball-and-socket (hip, shoulder): all planes.
- 3 Ligaments limit range of motion.** A forced movement beyond the normal range stretches or tears the ligament: that is a sprain. Pushed further, it displaces the joint: that is a dislocation.
- 4 Muscles and tendons pull, they do not push.** Muscle is elastic and highly vascularized; it heals well. Tendon is rich in nerves (intense pain) but poor in vessels (slow healing). Strain: overstretching. Tendinitis: repeated overload.
- 5 The vocabulary of movement describes the injury.** Flexion, extension, abduction, adduction, rotation, inversion and eversion. "Fall on a hyperextended hand" already points to the structure involved.

LEARNER'S MANUAL

Section 9.1, Anatomy and Physiology of the Locomotor System · Online encyclopedia

FIELD MANUAL

N/A

EN LIGNE

ThingLink scene: surface anatomy; musculoskeletal system video

02

Assessment and management

A single approach for fractures, dislocations, sprains and strains

Same approach, four injuries. Mechanism, PQRST, comparative exam, CSM, then RICE, immobilization and the evacuation decision.

Situation	Management	Evacuation
Closed fracture, CSM intact	RICE, immobilize, CSM every 15 min	Semi-urgent
Closed fracture, CSM impaired	Gentle in-line traction if angulated long bone, immobilize, shock	Urgent
Open fracture	Bleeding, irrigation without scrubbing the bone, dressing, splint, access to the wound	Urgent
Dislocation	Position found, CSM; gentle reduction only for finger, toe, kneecap	Urgent if CSM impaired, otherwise prompt
Sprain, strain	RICE, support, rest	Depending on ability to walk

Figure 2. From the situation found to the speed of evacuation.

- 1 The mechanism guides everything.** Direct force (fracture at the point of impact), indirect (at a distance) or twisting (sprain, spiral fracture). Type, direction and magnitude of the force: your first objective data.
- 2 PQRST, then comparative exam.** One limb at a time, palpation along the full length of the bone, comparison with the uninjured side, active then passive range of motion. Rings and watches removed. When in doubt, treat as a fracture.
- 3 CSM: the finding that decides the evacuation.** Circulation (distal pulse, colour, warmth, capillary refill), sensation, motion. Impaired CSM makes the evacuation urgent and justifies, for a grossly angulated long bone, gentle in-line traction. Stop if there is resistance or increasing pain.
- 4 RICE, and what not to do.** Ice 15 to 20 minutes while monitoring CSM, rest, elevation, light non-restrictive compression. Heat and massage later, never on a contusion. Acetaminophen or NSAIDs for pain and swelling.
- 5 Dislocations: recognize, immobilize, do not play surgeon.** Shoulder: arm held away from the body, bulge at the front; immobilize without reducing, evacuate quickly. Finger, toe, kneecap without suspected fracture: gentle reduction possible, then CSM. Hip and knee: advanced care only.

03

Splints

The immobilization principles you will apply in class

Immobilizing means preventing unnecessary movement to relieve pain, relax the muscles, protect vessels and nerves, and keep a closed fracture from becoming open during transport.

After any immobilization: CSM.



Figure 3. Four steps, in this order, whatever the splint.

- 1 What to immobilize: the neighbours rule.** Fractured bone: the joint above and the joint below. Injured joint: the bone above and the bone below. For the lower limbs, the uninjured leg can serve as support; the knee stays flexed 5 to 10 degrees with padding under the hollow.
- 2 Three families of splints, plus improvisation.** Soft (sleeping bag, clothing, foam), rigid (wood, plastic, padded aluminum such as the SAM Splint), anatomical (finger to finger, leg to leg). Improvised: stick, paddle, pack frame, sleeping pad. The traction splint is reserved for the femur and contraindicated for joints.
- 3 The configurations to know.** Single, double, sugar-tong (immobilization in both planes), simple and tubular sling, figure-eight bandage for the clavicle. In class, the list is limited to: clavicle, humerus, elbow, forearm, wrist, knee, ankle, foot, lower leg.
- 4 Firm, never restrictive.** Adhesive tape keeps its compression; elastic bandages loosen or tighten: do not overtighten them. Leave an extremity accessible for CSM. In prolonged immobilization, release periodically.
- 5 In cold weather, the splint cools the limb.** An immobilized patient produces less heat and circulation slows in the limb: risk of hypothermia and frostbite. Insulate the limb, insulate the patient, monitor the colour and warmth of the extremities.

LEARNER'S MANUAL

Section 9.7, Immobilization · Online encyclopedia

FIELD MANUAL

Pages 128 to 154: look at the picture, reproduce it

IN CLASS

Workshop: one bone and one joint per participant, with your own equipment